

Tommaso Calò

✉ tomcal@itu.dk 🌐 tommasocalo.github.io 🎓 Google Scholar

Bio

I am a **Human-Computer Interaction** and **Artificial Intelligence** researcher, focusing on **end-user development** and **generation of user interfaces**. I am currently a Postdoctoral Researcher at the  **IT University of Copenhagen** and the  **Pioneer Centre for AI**. Previously, I completed my Ph.D. at the  **Politecnico di Torino** in the  **e-Lite research group** under the supervision of **Luigi De Russis**. During my Ph.D., I visited the **Teachable AI Lab** at the  **Georgia Institute of Technology** under the supervision of **Christopher MacLellan**.

EMPLOYMENT & RESEARCH AFFILIATIONS

- | | |
|--|--|
| Postdoctoral Researcher
<i>IT University of Copenhagen</i> | Mar 2026 – present
<i>Copenhagen, Denmark</i> |
| <ul style="list-style-type: none">Conducting research on practices in generative user interface design and on methods for improving generative UI systems. | |
| Affiliated Researcher
<i>Pioneer Centre for AI</i> | Mar 2026 – present
<i>Copenhagen, Denmark</i> |
| <ul style="list-style-type: none">Serving as an affiliated researcher at the Danish National Research Foundation flagship AI research center. | |
| Postdoctoral Researcher
<i>Polytechnic University of Turin</i> | Jul 2025 – Feb 2026
<i>Turin, Italy</i> |
| <ul style="list-style-type: none">Conducted research in AI-assisted interactive systems co-creation. | |
| Visiting Scholar
<i>Georgia Institute of Technology</i> | Feb 2024 – Jul 2024
<i>Atlanta, USA</i> |
| <ul style="list-style-type: none">Conducted research on Human-like Learning in AI Systems under Prof. Christopher MacLellan. | |

EDUCATION

- | | |
|---|--|
| Ph.D. in Artificial Intelligence
<i>Polytechnic University of Turin</i> | Nov 2021 – Jul 2025
<i>Turin, Italy</i> |
| <ul style="list-style-type: none">Dissertation: "From Human Representations to AI Realization: Algorithms and Tools for Creating and Refining Interactive Systems." Supervisors: Prof. Luigi De Russis, Prof. Fulvio Corno. | |
| M.Sc. in Data Science and Engineering
<i>Polytechnic University of Turin</i> | Sep 2019 – Oct 2021
<i>Turin, Italy</i> |
| <ul style="list-style-type: none">Thesis: "Financial Time Series Summarization." Supervisors: Prof. Luca Cagliero, Dr. Jacopo Fiori. | |
| B.Sc. in Management Engineering
<i>Polytechnic University of Turin</i> | Sep 2015 – Mar 2019
<i>Turin, Italy</i> |
| <ul style="list-style-type: none">Thesis: "Software Cost Optimization in an R&D Automotive Center (Marelli)." Supervisor: Prof. Riccardo Chiaberge. | |

TEACHING EXPERIENCE

- | | |
|---|---|
| MOOC "Human-AI Interaction" — Author and Instructor
<i>Politecnico di Torino</i> | Dec 2025 – Feb 2026
<i>Italy</i> |
| <ul style="list-style-type: none">Designed and instructed a public Massive Open Online Course on Human-AI Interaction, funded by the EU Digital Education Hub. | |
| Database — Contract Professor
<i>University of Eastern Piedmont</i> | Jan 2026 – Feb 2026
<i>Vercelli, Italy</i> |
| <ul style="list-style-type: none">Lectured on SQL and relational design. | |
| Human-AI Interaction — Co-Instructor
<i>Polytechnic University of Turin</i> | Jan 2026 – Feb 2026
<i>Turin, Italy</i> |
| <ul style="list-style-type: none">Lectured on Generative AI and UI design patterns to Ph.D. candidates. | |
| Challenge@PoliTo by Reply — Project Mentor
<i>Polytechnic University of Turin (industry collaboration with Reply)</i> | Sep 2025 – Jan 2026
<i>Turin, Italy</i> |
| <ul style="list-style-type: none">Mentored student teams on agentic solutions for industrial challenges proposed by Reply, an international consulting and digital services firm. | |

Python Programming for Aerospace — Lead Instructor

ITS Foundation

Sep 2023 – Feb 2026

Turin, Italy

- Designed and taught the full Python curriculum to vocational students in the Mechatronics track.

Human-Computer Interaction — Teaching Assistant

Polytechnic University of Turin

Sep 2022 – Feb 2026

Turin, Italy

- Guided students in Computer Engineering annually through the full design cycle, from user research to AI-driven prototyping.

Computer Vision — Teaching Assistant

Polytechnic University of Turin

Feb 2023 – Jul 2023

Turin, Italy

- Led 12 hands-on sessions for B.Sc. students on image feature extraction, optics, and deep learning methods.

HONORS, AWARDS & COMPETITIVE RECOGNITION

- European Commission Seal of Excellence** — Marie Skłodowska-Curie Postdoctoral Fellowships 2025. Project GENUIS, score 94.60/100.
- Best Late-Breaking Result Award** — ACM SIGCHI Symposium on Engineering Interactive Computing Systems (EICS), 2025. For "Bridging Web and Figma" (Russo, Calò, De Russis); co-supervisor and ideator.
- Outstanding Reviewer Recognition** — ACM Conference on Designing Interactive Systems (DIS), 2026.
- CINECA ISCRA-C Computational Grants** (×3) — Italian National Supercomputing Consortium, peer-reviewed competitive allocations. ~192,000 GPU-hours total (~€72,000 in equivalent compute value).
- Postdoctoral Fellowship** (×2) — Politecnico di Torino. Total ~€80,000.
- Ph.D. Scholarship** — Politecnico di Torino, 2021–2025. ~€120,000.

SELECTED RESEARCH PROJECTS

AccessCoVE

EU Commission Funded Project

Nov 2024 – Feb 2026

Turin, Italy

- Led the definition of AI-supported educational frameworks for inclusive participation.

AI-ALOE

NSF & Accenture Funded

Feb 2024 – Jul 2024

Atlanta, USA

- Contributed to the development of cognitive AI assistants for adult learning at Georgia Tech, within the NSF AI Institute for Adult Learning and Online Education.

PUBLICATIONS

JOURNAL ARTICLES

- T. Calò**, A. Sillano, L. De Russis. "MorphGUI: Real-Time GUIs Customization with Large Language Models." *International Journal of Human-Computer Studies (IJHCS)*, Elsevier, 2025. doi:10.1016/j.ijhcs.2025.103695
- T. Calò**, L. De Russis. "Advancing Code Generation from Visual Designs through Transformer-Based Architectures and Specialized Datasets." *Proceedings of the ACM on Human-Computer Interaction (PACM-HCI)*, 2025. doi:10.1145/3734190
- T. Calò**, L. De Russis. "Enhancing Smart Home Interaction through Multimodal Command Disambiguation." *Personal and Ubiquitous Computing (PUC)*, Springer, 2024. doi:10.1007/s00779-024-01827-3

CONFERENCES AND SYMPOSIA

- T. Calò** et al. "Measuring the Semantic Accessibility Gap in LLM-Generated Web UIs." *ACM CHI Conference on Human Factors in Computing Systems*, 2026.
- T. Calò** et al. "WebUI-95: A Large-Scale Dataset of Normalized Web Interfaces via UI-to-Code Generation." *ACM CHI Conference on Human Factors in Computing Systems*, 2026.
- T. Calò** et al. "CAT-AI: Supporting Teacher Workflows with AI-Assisted Exercise Creation." *ACM CHI Conference on Human Factors in Computing Systems*, 2026.
- A. Sillano, **T. Calò**, L. De Russis. "From Text-First to Structure-First: A Visual Pipeline for Narrative Writing." *ACM CHI Conference on Human Factors in Computing Systems*, 2026.
- A. M. Roffarello, **T. Calò**, L. De Russis. "Investigating How Computer Science Researchers Design Their Co-Writing Experiences With AI." *ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2025. doi:10.1145/3706598.3713205
- T. Calò**, L. De Russis. "DeepFlow: A Flow-Based Visual Programming Tool for Deep Learning Development." *ACM Conference on Intelligent User Interfaces (IUI)*, 2025. doi:10.1145/3708359.3712109

10. A. Gupta, J. Reddig, **T. Calò**, D. Weitekamp, C. J. MacLellan. "Beyond Final Answers: Evaluating Large Language Models for Math Tutoring." *International Conference on Artificial Intelligence in Education (AIED)*, 2025.
11. F. Russo, **T. Calò**, L. De Russis. "Towards Step-Aware ITSs: Generation and Evaluation of Synthetic Step-by-Step Exercise Solutions." *ACM Conference on Learning at Scale (L@S)*, 2025.
12. **T. Calò**, L. De Russis. "Evaluating Visual Prompting Modalities for Generative AI-Assisted UI Design." *International Symposium on End-User Development (IS-EUD)*, 2025. doi:10.1007/978-3-031-95452-8_11
13. F. Russo, **T. Calò**, L. De Russis. "Bridging Web and Figma: Automating Large-Scale UI Dataset Generation for AI-Enhanced Design." *ACM SIGCHI Symposium on Engineering Interactive Computing Systems (EICS)*, 2025. **Best Late-Breaking Result Award**. doi:10.1145/3731406.3734974
14. **T. Calò**, C. MacLellan. "Towards Educator-Driven Tutor Authoring: Generative AI Approaches for Creating Intelligent Tutor Interfaces." *ACM Conference on Learning at Scale (L@S)*, 2024. doi:10.1145/3657604.3664694
15. **T. Calò**, L. De Russis. "Leveraging Large Language Models for End-User Website Generation." *International Symposium on End-User Development (IS-EUD)*, 2023. doi:10.1007/978-3-031-34433-6_4
16. **T. Calò**, L. De Russis. "Towards a Visual Programming Tool to Create Deep Learning Models." *ACM SIGCHI Symposium on Engineering Interactive Computing Systems (EICS)*, 2023.
17. **T. Calò**. "Rethinking Design Workflows with Artificial Intelligence." *Doctoral Consortium @ CHIItaly*, 2023.
18. **T. Calò**, L. De Russis. "Style-Aware Sketch-to-Code Conversion for the Web." *ACM SIGCHI Symposium on Engineering Interactive Computing Systems (EICS)*, 2022.
19. **T. Calò**, L. De Russis. "Creating Dynamic Prototypes from Web Page Sketches." *ACM SIGPLAN Workshop on Programming Abstractions and Interactive Notations (PAINT)*, 2022.
20. J. Fior, L. Cagliero, **T. Calò**. "Generating Comparative Explanations of Financial Time Series." *European Conference on Advances in Databases and Information Systems (ADBIS)*, 2022.

DOCTORAL DISSERTATION

21. **T. Calò**. "From Human Representations to AI Realization: Algorithms and Tools for Creating and Refining Interactive Systems." Ph.D. Dissertation, Politecnico di Torino, 2025.

INVITED RESEARCH TALKS

- **Eindhoven University of Technology (TU/e)** — "You Can't Always Get What You (Prompt): From Specification to Observation in Generative Interfaces" — invited by Prof. Minha Lee — May 2026.
- **IT University of Copenhagen** — Lunch Seminar — "Who Pays for the Free Lunch in Generative UI?" — May 2026.
- **Microsoft Research** — "You Can't Always Get What You (Prompt): From Specification to Observation in Generative Interfaces" — invited by Sian Lindley — February 2026.
- **University of Bologna** — "Human-Centered AI" — invited by Prof. Catia Prandi — November 2025.
- **Worcester Polytechnic Institute (WPI)** — "End-User Development for User Interfaces" — invited by Prof. Neil Heffernan — July 2024.

SERVICE & LEADERSHIP

EDITORIAL / JUDGING ROLES

- **Associate Chair**: **ACM CHI** 2025 (Late-Breaking Work), **ACM IUI** 2026, **ACM AVI** 2026.
- **Senior Program Committee**: **ACM CHI** 2025 (Late-Breaking Work), **ACM IUI** 2026.

CONFERENCE ORGANIZING ROLES

- **Tutorials Chair** — ACM Interactive Surfaces and Spaces (ISS), Turin, 2026.
- **Social Media Chair** — European Conference on Computer-Supported Cooperative Work (ECSCW), 2024.
- **Social Media Chair** — ACM SIGCHI Symposium on Engineering Interactive Computing Systems (EICS), 2024.

PROGRAM COMMITTEE / PEER REVIEW

- **ACM CHI**, **ACM IUI**, **ACM EICS**, **ACM DIS**, **IS-EUD**, **PACM-HCI**, **ACM L@S**.

OTHER PROFESSIONAL AFFILIATIONS

- **ACM** — Association for Computing Machinery, member.
- **ACM SIGCHI** — Special Interest Group on Computer-Human Interaction, member.

PH.D. STUDENT CO-SUPERVISION

- **Giuseppe Arbore** — co-supervised with Prof. Luigi De Russis, Politecnico di Torino.

M.Sc. THESIS CO-SUPERVISION

- **Lorenzo Cuccu** — *CAT-AI: An LLM-Based System for Creating Customized Didactical Exercises*
- **Simone Santarcangelo** — *Automatic Generation of Custom Web Prototypes from Sketches*
- **Francesca Russo** — *GUI Representation Learning for Downstream Real-World Applications*
- **Filippo Castellarin** — *DeepBlocks: Visually Constructing, Debugging and Training Deep Neural Networks*
- **Andrea Sillano** — *Generation of Tailored Graphical User Interfaces via Artificial Intelligence*
- **Ivan Dario Contreras Perez** — *Large Language Models for Ambiguity Detection and Resolution in Smart Homes*
- **Giuseppe Salvi** — *Web UI Code Generation: A Transformer-Based Model Applied to Real-World Screenshots*